一、RC 串联电路暂态过程

$$R_{测}=99.80\Omega，C_{测}=99.61nF，T_{\frac{1}{2}测}=6.6\mu s$$

$T_{\frac{1}{2}}=Tln2=R_{测}C_{测}ln2=6.89\mu s$，相对误差$E=\frac{|T_{\frac{1}{2}测}-T_{\frac{1}{2}}|}{T_{\frac{1}{2}}}=4.21\%<5\%$

二、RL 串联电路暂态过程

$$R_{测}=99.80\Omega，L=0.01H，T_{\frac{1}{2}测}=68.0\mu s$$

$T_{\frac{1}{2}}=Tln2=\frac{Lln2}{R_{测}}=69.44\mu s$，相对误差$E=\frac{|T_{\frac{1}{2}测}-T_{\frac{1}{2}}|}{T_{\frac{1}{2}}}=2.07\%<5\%$

三、RLC 串联电路暂态过程

$L=0.01H$，$C_{测}=99.61nF$，临界阻尼时$T_{0.264测}=31\mu s$

$\beta=\frac{R}{2L}$，$\omega_0=\frac{1}{\sqrt{LC}}$，临界阻尼时$\beta=\omega_0$，即$R=2\sqrt{\frac{L}{C}}$

$T_{0.264}=\frac{1}{\beta}=\sqrt{LC}=31.56\mu s$，相对误差$E=\frac{|T_{0.264测}-T_{0.264}|}{T_{0.264}}=1.77\%<5\%$

四、RLC 谐振电路稳态过程

$R_{测}=99.80\Omega$，$L=0.01H$，$C_{测}=99.61nF$，$f_{0测}=5.023kHz$，$U_0=4.0V$，通频带（$\frac{U}{\sqrt{2}}$处）宽度对应频率

$f_1=5.915kHz$，$f_2=4.243kHz$

由φ = 2πfΔt, 整理实验记录数据可得以下表格：

表 1 RLC 谐振电路稳态过程数据处理

		$U_{0.94}$	$U_{0.9}$	$U_{0.8}$	$U_{0.7}$	$U_{0.6}$	$U_{0.5}$	$U_{0.3}$	$U_{0.1}$
	U/V	3.76	3.60	3.2	2.8	2.4	2.0	1.2	0.4
$f<f_0$	f/kHz	4.753	4.663	4.453	4.253	4.003	3.703	2.813	0.803
	Δt/μs	-14.6	-16.6	-22.8	-29.8	-37.4	-48.0	-75.2	-298
	φ/rad	-0.436	-0.486	-0.638	-0.796	-0.941	-1.117	-1.329	-1.504
$f>f_0$	f/kHz	5.286	5.363	5.643	5.903	6.313	6.743	8.713	21.413
	Δt/μs	9.8	12.4	18.4	21.3	24.0	26.4	24.8	13.4
	φ/rad	0.325	0.418	0.652	0.790	0.952	1.119	1.358	1.803

$f_{0理论}=\frac{1}{2\pi\sqrt{LC}}=5.028kHz$，相对误差$E_1=\frac{|f_{0测}-f_{0理论}|}{f_{0理论}}=0.99\%<5\%$

$Q_{理论}=\frac{1}{R}\sqrt{\frac{L}{C}}=3.074$，$Q_{测}=\frac{f_{0测}}{f_1-f_2}=3.004$，误差分析$E_2=\frac{|Q_{测}-Q_{理论}|}{Q_{理论}}=2.28\%<5\%$

理想相移 $\phi=\arctan(\frac{2\pi fL-\frac{1}{2\pi fC}}{R})$

幅值-频率曲线和相移-频率曲线见下图。

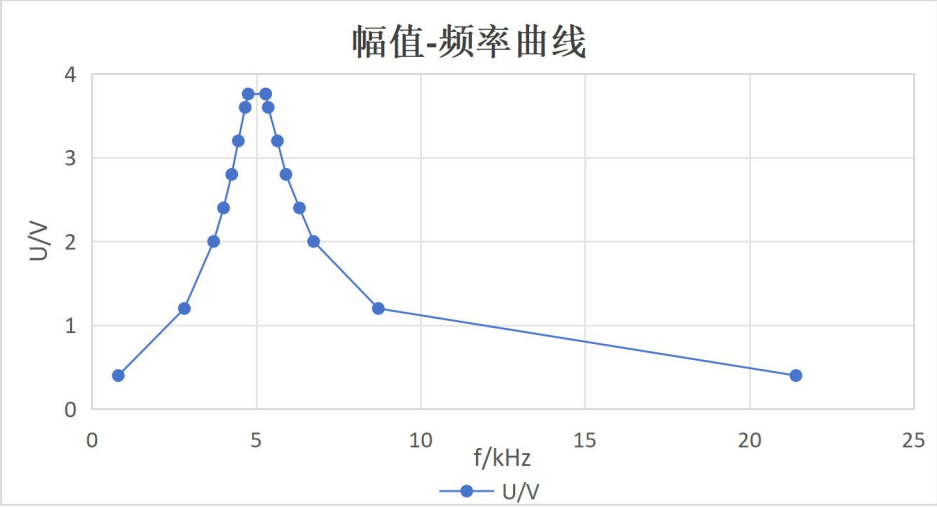


图 1 幅值-频率曲线

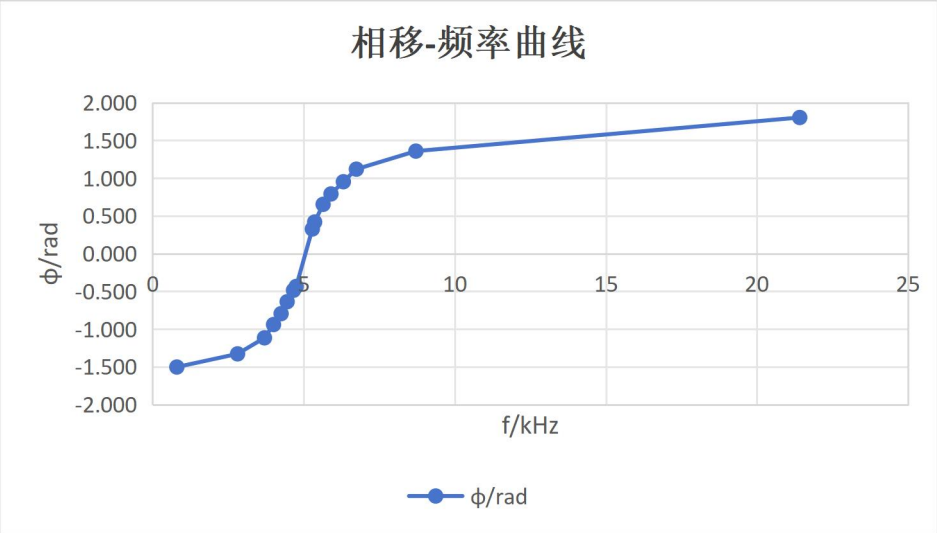


图 2 相移-频率曲线

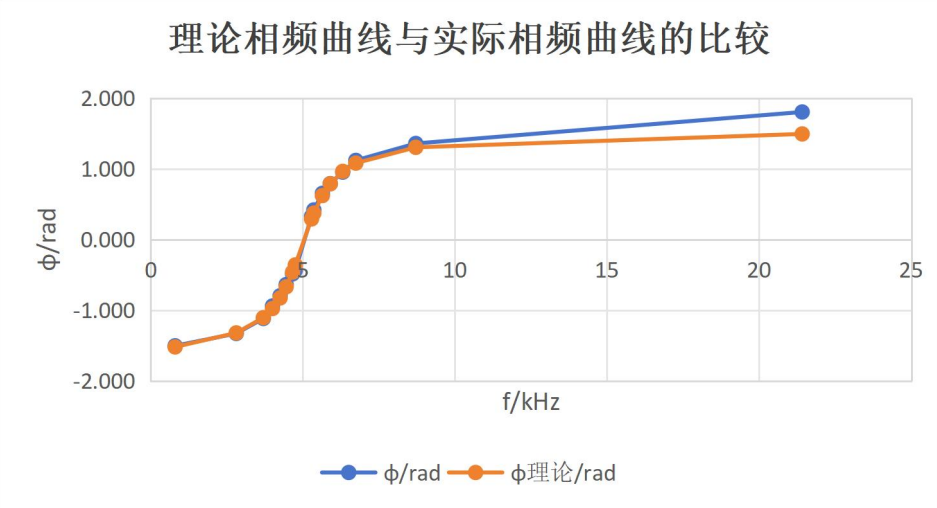


图 3 理论相频曲线与实际相频曲线的比较

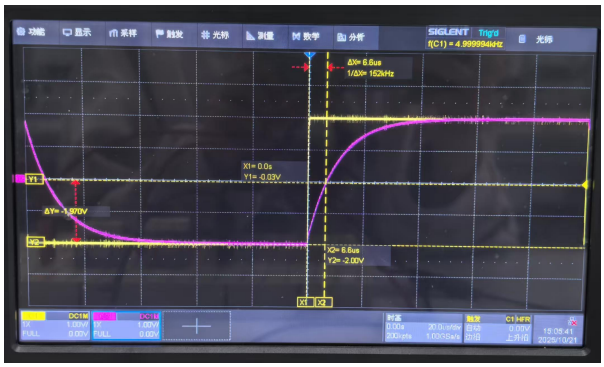


图 4 RC 串联电路波形图

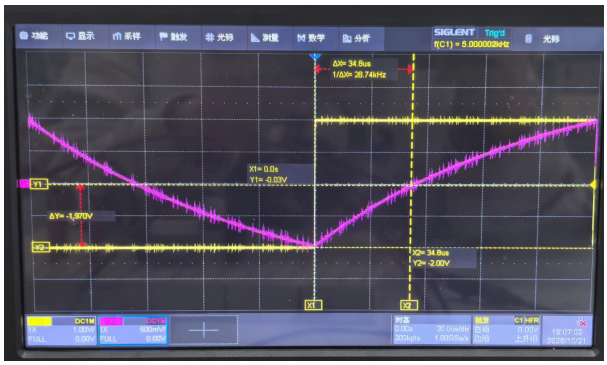


图 5 RL 串联电路波形图



图 6 RLC 串联电路波形图（过阻尼）

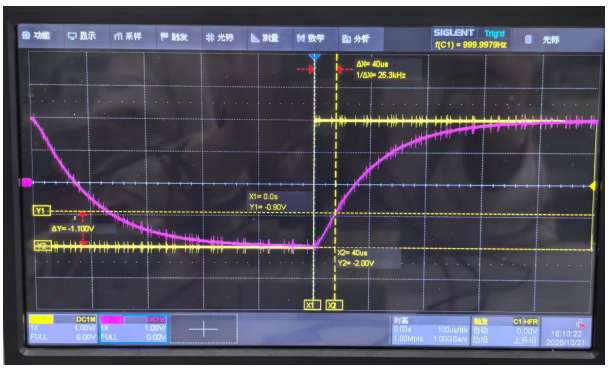


图 7 RLC 串联电路波形图（欠阻尼）

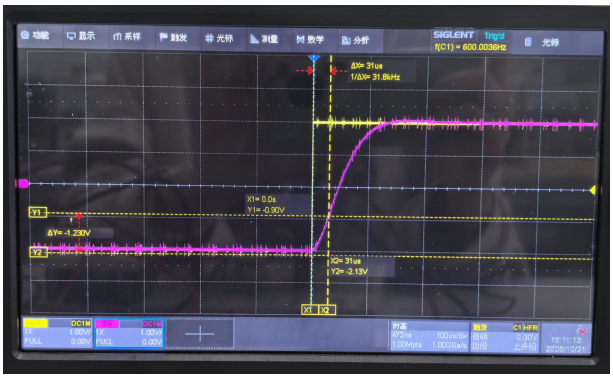


图 7 RLC 串联电路波形图（临界阻尼）